

This question paper contains 8 printed pages]

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Integrated M.C.A. (First Semester)

Examination, Dec. 2022

IMCA-103

DIGITAL ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

*This question paper contains three
sections as under :*

Section-A

Max. Marks-10

*This section shall have one compulsory question
comprising 10 questions (maximum 20 words answer)
taking two questions from each unit. Each question shall
be of one mark and total marks of this section will
be 10. This section will be compulsory in the paper.*

Section-B

Max. Marks-25

This section will carry **25** marks with equally divided into five long answer type questions (answer about in **250** words) and students are instructed to attempt **five** questions by selecting one question from each unit.

Section-C

Max. Marks-35

Will carry **35** marks with **five** long answer type questions comprising one compulsory question (Question No. 7) of **15** marks and four questions of **10** marks each. Students are instructed to attempt total **three** questions with **one** compulsory question (answer about in **500** words) and any **two** more questions (answer about in **400** words) out of remaining four questions.

Section A

1. (i) How signed numbers are represented ? Give suitable example.
- (ii) Convert binary number 111000 into decimal number.

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- (iii) What is logic gates ? Draw symbols of NAND, NOR and EX-OR gates.

- (iv) Write Demorgan's theorem and show its truth tables.

- (v) Draw Karnaugh map for the logical expression :

$$Y = \bar{A} \bar{B} \bar{C} \bar{D} + A \bar{B} D + A B \bar{D} + C D$$

- (vi) Implement a half adder circuit using NAND gate only.

- (vii) What is SR Latch ? Draw its logic circuit using NOR OR NAND gate. Also give its truth table.

- (viii) Which type of flip-flops are used in counters ? Enlist the applications of counters.

- (ix) Why are A/D and D/A converters necessary ? Explain the principle of D/A converter generally used.

- (x) Enlist the performance characteristics of A/D converters. Also write the formula for percentage resolution (if the number of bits is N).

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Section B

UNIT-I

2. State the advantages of binary system. Convert the following : 5

(i) $(11.0101)_2 = (?)_{10}$

(ii) $(0.6875)_{10} = ()_2$

(iii) $(1101.11011)_2 = ()_8$

(iv) $(123 F)_{16} = ()_{10}$

Or

Explain gray code and Hamming code with example. Convert each binary number to gray code : 5

(i) $(11111)_2$

(ii) $(1001000)_2$

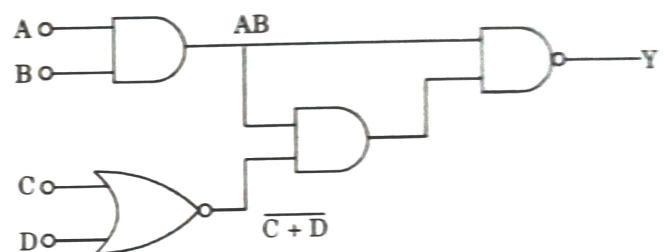
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UNIT-II

3. Write the rules of Boolean algebra, use Demorgan's theorem and Boolean algebra laws to simplify the circuit shown in figure : 5



Or

Discuss the various logic gates with their logic symbol and truth table. Prove the following using Boolean algebra rules : 5

(i) $A + \bar{A}B = A + B$

(ii) $(A + B)(A + C) = A + BC$

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UNIT-III

4. Explain functioning of encoder with its applications. Give logic diagram and truth table of decimal to BCD encoder. 5

Or

Explain the functioning of multiplexer with its application. Give Boolean expression and truth table of eight input multiplexer. 5

UNIT-IV

5. Discuss the different types of counters. Also give their application. 5

Or

Explain edge triggered JK flip-flop with truth table and waveforms. Also discuss the race condition in JK flip-flop. 5

UNIT-V

6. Describe the weighted resistor network and binary ladder network for D/A conversion. Also give performance characteristics of D/A converters. 5

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Or

Describe the dual slope integrator and successive approximation methods for A/D conversion. Also give performance characteristics of A/D converters. 5

Section C

7. (a) If $A = -91$ and $B = +46$, express A and B in 2's complement form and find $A + B$. Express result in 8 bit 2's complement form.

(b) Discuss the utility of an inverter gate. Write its truth table.

(c) What is meant by don't care conditions in k-map ? Give example.

(d) Differentiate between combinational circuit and sequential circuit.

(e) Discuss the term quantization error in analog to digital (A/D) converter. $5 \times 3 = 15$

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8. Why is a code necessary ? Discuss. Perform the following operation : 10

(i) BCD addition of $(0110 + 0111)$

(ii) 2's complement subtraction of $(1000 - 1100)$

(iii) Multiply $(10.101)_2$ by $(0.101)_2$

(iv) Convert $(177.25)_{10}$ to octal number.

9. Showing the universal property, implement the different gate using NOR gates. Also implement the expression $Y = AB + CD$ using only NAND gates. 10

10. Differentiate between the following : 10

(i) Half adder and full Adder

(ii) Multiplexer and demultiplexer

(iii) POS and SOP standard form of Boolean expression

(iv) Encoder and Decoder.

11. (i) Briefly explain different flip-flop parameters and also give applications of flip-flop.

(ii) Discuss the different type of Shift Registers in brief. 5+5=10